



The scam shield for **the people you love.**

A Chrome extension that catches scams in real time — and explains the danger in words a 75-year-old actually understands.

Live: safesurf-blush.vercel.app · **Code:** github.com/prokiller67-crypto/safesurf

Brainwave 2026 · AI × Cybersecurity · Nikita Nizhevyasov

— THE PROBLEM

Online fraud is a tax on old age.

Scammers deliberately target people who grew up before the internet — with fake virus pop-ups, “grandchild in trouble” messages and pixel-perfect bank clones.

\$4.8B

lost by Americans aged 60+ to online fraud in 2024 (FBI IC3)

147k

complaints from seniors in one year
— the fastest-growing victim group

×4

seniors lose ~4× more per incident than victims under 30

Detection isn't the gap. Communication is.

What security tools say

NET::ERR_CERT_AUTHORITY_INVALID

Deceptive site ahead. Attackers
may trick you into doing
something dangerous.

Jargon → fear → “I’ll just click Continue”

What SafeSurf says

🔴 This page pretends to be your bank.

👉 Close it and don't type your
password. If unsure, call your family.

Plain words → understanding → right action

— THE SOLUTION

A guardian that speaks human.

SafeSurf watches every page and shows a big, high-contrast traffic-light banner the moment something is wrong — with one plain-language explanation and one concrete action.

- ✓  **Danger** /  **Caution** /  **silence** — no nagging on safe pages, loud when it matters
- ✓ **Huge type, high contrast** — designed for older eyes
- ✓ **One action** — “Close this page”, not a settings menu
- ✓ **Zero jargon** — “fake website”, never “phishing domain”
- ✓ **Body font: Atkinson Hyperlegible** — a typeface built for low-vision readers
- ✓ **Works without setup** — heuristics protect even with no API key

Two layers. Fail-safe by design.

LAYER 1 · INSTANT · OFFLINE

Local heuristics

- Typosquatting: `paypal.com`, look-alike unicode
- Brand-in-address tricks: `paypal.com.evil.xyz`
- 12 scam-script patterns: fake prizes, fake virus alerts, gift-card payments, remote access
- Dark patterns: countdowns, “do not close”



LAYER 2 · CLAUDE AI

AI second opinion

- Fires **only** when Layer 1 flags something
- Weighs URL + page text + heuristic evidence
- Structured JSON verdict: risk, explanation, advice
- Written for a senior, decisive by prompt design



VERDICT

The banner

- Heuristic warning appears in <50 ms
- AI upgrades it seconds later
- API down? Heuristic warning stays
- Safe page? Total silence

Small, honest, production-shaped.

- ✓ **Chrome Manifest V3** — TypeScript, esbuild, content script + service worker
- ✓ **Claude Opus 4.8** via official SDK with **structured outputs** (`output_config.format`) — the verdict is schema-validated JSON, never free text
- ✓ **Prompt-injection hardened** — page text is fenced as untrusted content
- ✓ **Privacy-first** — page text leaves the machine *only* when heuristics flag it; normal browsing is never sent anywhere
- ✓ **Zero-setup hosted mode** — Vercel proxy holds the key; rate-limited, capped, swappable for BYOK
- ✓ **Tested** — heuristic engine verified against scam & legit URL corpus

Watch it catch three real scam patterns.

windows-defender-alert.top

 **DANGER — this looks like a scam**
This page pretends to be a Microsoft virus warning.

Fake “5 viruses found”, phone number, remote-access ask.

win-your-iphone.click

 **DANGER — this looks like a scam**
Nobody gives away free iPhones. The timer is fake.

Prize scam with countdown + card form.

paypal-verify.top/login

 **DANGER — this looks like a scam**
This is not the real PayPal website.

Look-alike bank login hunting for passwords.

► Live demos: safesurf-blush.vercel.app — install the extension, click a tile, watch the banner.

— IMPACT

Built for the most-targeted, least-protected users online.

For seniors

Independence without fear. They keep banking, shopping and video-calling — SafeSurf only speaks up when something is genuinely wrong.

For families

Install once during a Sunday visit. Every warning ends with a human fallback: “if unsure, call your family.”

For everyone

Plain-language security helps kids, non-native speakers and cognitively-impaired users too. Accessibility is the product, not a feature.

From one banner to a family safety net.

NOW — MVP

- Chrome extension, 2-layer detection
- Open source · hosted AI works out of the box

NEXT 3 MONTHS

- Family Guardian dashboard: gentle alerts to a trusted contact on “danger” events
- Firefox / Edge ports
- Multilingual verdicts (ES, RU, HI...)

12 MONTHS

- Email & SMS scam scanning
- On-device small model for layer 2 — zero-cloud mode
- Partnerships: banks, senior-care orgs, ISPs

Unit economics scale: heuristics are free; AI calls happen only on flagged pages (<2% of browsing), pennies per user per month.

— CLOSING

Scammers weaponized language. We weaponized it back.

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 **SafeSurf** — Brainwave 2026